

Joe Lopez

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WORK EXPERIENCE

Microsoft

Redmond, WA

Sep 2021 – May 2025

Firmware Development Engineer

- Developed UEFI firmware for PCIe hardware initialization on company's first purpose-built ARM server platform, Cobalt 100. Ported open source libraries and contributed fixes upstream.
- Mitigated boot issues during chip and platform bringup by diagnosing ACPI issues using Windows Debugger (WinDbg) and ACPI tools, releasing a firmware patch in a timely manner to unblock others.
- Led the integration of firmware unit test suite into platform codebase, consuming open source framework and modifying it for our usecase. Worked with internal open-source owners to upstream enhancements for consumption by all first-party and third-party UEFI teams.
- Leveraged internally developed tool and agentic AI to generate firmware unit tests for previously uncovered lines of code using techniques such as RAG and multi-model approaches to achieve higher efficiency and scale to larger code files.

Robert Bosch Tool Corporation

Mount Prospect, IL

May 2019 – Aug 2019

Electrical Engineering Intern, Dremel

- Collaborated with Electrical and Mechanical engineers to create prototype tool with USB (UART) interface to decrease reliance on bluetooth chips
- Improved data acquisition capacity by rewriting firmware for custom tool logger
- Assembled and diagnosed issues on prototype tools using oscilloscope, multimeter

EDUCATION

University of Illinois at Urbana-Champaign

Aug 2017 – Jun 2021

- Bachelor's in Computer Engineering
- Theta Tau Professional Engineering Fraternity
 - Projects Chair 2019-2020
- Illinois Formula Electric
 - Data Acquisition Lead 2019-2020